

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain briefly the effect of crystal defect on material.
- Q.24 Explain the various types of cast iron with uses & properties.
- Q.25 What is heat treatment furnace? What are its different types? Explain any two.

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4th Sem / Automobile, Mechanical, Mech(Tool & Die)
Subject : Materials and Metallurgy

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is an alloy.
- a) cast iron b) steel
- c) brass d) all of above
- Q.2 In H.C.P. structure, the number of atoms in a unit cell are
- a) 1 b) 2
- c) 4 d) 6
- Q.3 Plastic deformation may take place due to
- a) slip only b) twinning only
- c) slip or twinning d) none of them

Q.4 Pearlite is a combination of

- a) Ferrite & Iron graphite
- b) Ferrite & Austenite
- c) Ferrite & Cementite
- d) None of the above

Q.5 Which of the following is/are Point defects?

- a) Schottky defect b) Vacancy defect
- c) Substitutional defect d) All of these

Q.6 The ability of a material to resist fracture due to high impact load is:

- a) Toughness b) Hardness
- c) Brittleness d) Malleability

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 HSS stands for _____

Q.8 The line above which the alloy is liquid is called _____

Q.9 Define ferrous metals.

Q.10 _____ Carbon steel contains carbon 0.3% to 0.8%.

Q.11 Define Refractory material.

Q.12 Define atomic radius.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Differentiate between slip and twinning.

Q.14 Give the classifications of materials.

Q.15 What are semi-conducting materials? What are its different types?

Q.16 What is a refractory material? What are its different types? Explain.

Q.17 What are line defects? What its types? Explain.

Q.18 Explain any four mechanical properties of materials.

Q.19 What is pyrometer? What are its different types.

Q.20 Differentiate between ferrous and non-ferrous metals.

Q.21 Explain T.T.T diagram.

Q.22 Write short note on Cyaniding.